

CHIRANJEEVI GOLI

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with experience in Python, engineering data analysis, and automation using Python, CAPL, and Vector CANoe through a Graduate Apprenticeship at Volvo Group. Built AI-powered projects including a RAG knowledge assistant and a natural language-to-SQL application. Passionate about applying data and AI to solve real-world problems through continuous learning and collaboration.

TECHNICAL SKILLS

Programming & Data Analysis: Python, SQL, Pandas, NumPy, Matplotlib, Statistical Analysis, C, CAPL

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Embeddings, ChromaDB, Machine Learning Fundamentals

Databases & Backend Development: FastAPI, MySQL, SQLAlchemy, REST APIs

Engineering Domain Knowledge: Basic Electronics, CAN, LIN, TCP/IP, DBC/LDF Configuration, ECUs

Tools & Platforms: Git, GitHub, VS Code, Postman, Excel, Vector CANoe, Jira, ET tool

AI Tools: Claude, ChatGPT, Google AI Studio

Soft Skills: Adaptability, Attention to Detail, Problem Solving, Ownership & Accountability, Learn-by-Doing

WORK EXPERIENCE

Graduate Apprentice Trainee – Data Automation & Test Engineering

May 2025 – May 2026

Volvo Group India Pvt. Ltd.

Data Pipeline Development & Test Automation

- Delivered CAPL-based automation supporting PMSL test execution alongside CANoe/CAPL and Python, improving repeatability and reducing manual intervention across multiple test scenarios.
- Built Python utilities to parse Excel-based test specifications, perform engineering unit conversions, and auto-generate CAPL function calls, reducing manual scripting time by 90%.
- Developed CAPL scripts and control panels for signal access and automated logging across thermal chambers, alternators, and DC regulated power supplies, simplifying data collection and analysis.
- Designed and configured Vector CANoe test environments integrating T2 and T3 hardware platforms, supporting reliable, repeatable laboratory validation.

Data Acquisition, Network Communication & Hardware Configuration

- Maintained CAN network configuration data by creating and correcting DBC files and CAN matrix content, with every change validated through structured technical review.
- Configured and validated LIN communication for alternator networks, including LDF setup and wiring, and verified combined CAN/LIN interaction scenarios.
- Independently researched and implemented TCP/IP socket communication in Vector CANoe using CAPL, enabling integration with an external DC regulated power supply for automated testing.
- Developed a Python utility that converted Excel-based test configurations into Neware XML files, eliminating manual channel setup and enabling simultaneous execution across 8 test channels.
- Flashed software onto T2 and T3 ECU variants to prepare test environments for validation.

Lab Operations & Documentation

- Operated and maintained laboratory equipment – multimeters, oscilloscopes, programmable power supplies, and electronic loads – to collect accurate performance data for analysis.
- Prepared technical documentation for developed automation tools and maintained lab equipment checklists to track operational status and readiness.

PROJECTS

FleetMind AI – RAG-Based Knowledge Agent for Heavy-Duty Truck Maintenance

Python, RAG, ChromaDB, Gemini API

- Designed and built an interactive Retrieval-Augmented Generation (RAG) assistant with Python, Streamlit, ChromaDB, Sentence Transformers, and the Google Gemini API to support fleet maintenance troubleshooting and predictive-maintenance decisions over heavy-duty vehicle manuals.
- Engineered an agentic retrieval-routing layer that evaluated retrieval confidence and triggered clarification prompts for ambiguous queries, improving answer reliability and reducing incorrect responses.
- Built a document ingestion pipeline with PyMuPDF and LangChain that extracted, chunked, embedded, and indexed 1,212 pages into 3,795 semantic chunks, enabling fast semantic search over technical fleet documentation.

E-Commerce Order Analytics – Data Cleaning, NumPy Performance & Visualization

Python, Pandas, NumPy, Matplotlib

- Cleaned and merged a messy, multi-table e-commerce dataset (~300,000 orders) in Pandas, fixing missing values, duplicate records, and mixed date formats, and diagnosed a real merge-explosion bug caused by duplicate join keys.
- Applied core NumPy operations – vectorization, boolean filtering, and broadcasting – to the order data and benchmarked a 23x speedup over plain Python loops on large arrays.
- Used Matplotlib to visualize revenue trends by category and country from grouped and pivoted data, turning cleaned datasets into clear, presentable insights.

AI Data Analyst Assistant

Python, Streamlit, Gemini API, SQLAlchemy, MySQL

- Built an interactive data analyst assistant with Streamlit and the Google Gemini API that lets users upload any CSV dataset and ask analytical questions in plain English.
- Implemented automated schema detection and natural-language-to-SQL generation, executing queries against a MySQL database via SQLAlchemy and returning clear, plain-text answers.
- Added automatic chart generation for trend and comparison questions plus context-aware follow-up handling, so it retains prior questions within a session.

Employee Management System

Python, FastAPI, MySQL, SQLAlchemy

- Built a REST API-based Employee Management System with FastAPI, SQLAlchemy, and MySQL, supporting complete CRUD operations end to end.
- Designed database models and implemented request validation with Pydantic, using dependency-based database session management for reliable, testable endpoints.

EDUCATION

B.Tech, Electronics and Communication Engineering

Vasireddy Venkatadri Institute of Technology, Guntur

2020 – 2024

CGPA: 8.92

CERTIFICATIONS

- **Python Programming** – Coursera